

ODD-OUTDEGREE TRANSPOSE DYNAMICS ON LOOPED DIGRAPHS: PARITY COLLAPSE, FIBRES, AND COMPONENTS

ANONYMOUS

ABSTRACT. On a labelled looped digraph, reverse every arc and then toggle every ordered pair, loops included, inside the set of vertices of odd current outdegree. Equivalently, on $M_n(\mathbb{F}_2)$ we iterate $\Phi(A) = A^\top + (A\mathbf{1})(A\mathbf{1})^\top$. We prove that one step maps onto the even-total-parity hyperplane, which is the complete recurrent set, and that all recurrent periods divide four. Every codomain fibre has size 0, 1, or $2^{n-1} + 1$, according to an explicit column-parity test. We then give exact formulas for fixed points, two- and four-cycles, depth-one states, and the finite dynamical zeta function. A second factorisation views the state-dependent left factor as the identity or an involutory transvection, versus a singular projection, and independently recovers the second and fourth iterates. Static graph complementation, binary-margin counting, and finite-map zeta bookkeeping are treated as background. The owner search is bounded; no novelty, priority, or external-release claim is made.

1. THE UPDATE AND ITS SUBTRACTION BOUNDARY

Fix $n \geq 1$, work over $\mathbb{F}_2$, and let $\mathbf{1}$ be the all-one column. A matrix $A \in M_n(\mathbb{F}_2)$ is the adjacency matrix of a labelled directed graph in which loops are allowed. Put

$$r = A\mathbf{1}, \quad c = A^\top \mathbf{1}, \quad \tau = \mathbf{1}^\top A\mathbf{1}.$$

Thus r and c are the out- and indegree parity vectors and $\tau = \mathbf{1}^\top r = \mathbf{1}^\top c$ is total arc parity. Define

$$(1) \quad \Phi(A) = A^\top + rr^\top.$$

The second summand toggles the complete looped subdigraph on the current odd-outdegree vertices. It is a rank-*at-most-one* correction and is recomputed after every step.

Local complementation and its binary-matrix formulation are classical [2, 7]; pivot and loop complementation have an established algebraic calculus [3]. Static subgraph complementation, including its rank and algorithmic interfaces, is also a separate mature operation [4, 6]; see also [arXiv:2502.15675](#). Modular invariant rings for transpose group actions on full 2×2 matrix spaces form another neighboring, but static, interface [5]. All such static operations and invariant rings, transpose, outer products, transvections, binary margin enumeration, and generic functional-graph bookkeeping receive zero contribution credit here. The residual question is only the exact temporal structure of the state-dependent update (1).

The internal subtraction is equally strict. P102's involution-norm dynamics and P103's double-adjugation dynamics receive zero credit. P125 acts on pairs in a nonsingular quadratic space, has a quadratic/polar three-bit quotient, and permits tail two and period three. In contrast, the present carrier is all of $M_n(\mathbb{F}_2)$, its feasible row/column-margin quotient has dimension $2n - 1$, its tail is at most one, and its recurrent periods are only 1, 2, 4. Thus the literal systems are not conjugate; generic quotient, matrix, quadratic-map, cycle, fibre, component, and zeta packaging still receive zero credit. A bounded formula-and-owner search found no literal-map match; a bounded non-hit is not a novelty certificate.

2. THE PARITY QUOTIENT AND ALL ORBITS

For $B = \Phi(A)$, write its margins as $r(B), c(B), \tau(B)$.

Lemma 2.1 (quotient law). *For every A ,*

$$(2) \quad r(B) = c + \tau r, \quad c(B) = (1 + \tau)r, \quad \tau(B) = 0.$$

Consequently the even-total hyperplane $\mathcal{H}_n = \{A : \tau(A) = 0\}$ is invariant; on it the margin pair is swapped.

Proof. Multiplying (1) on the right by $\mathbf{1}$ gives $r(B) = c + r(r^\top \mathbf{1}) = c + \tau r$. Transposing first gives $c(B) = r + r(r^\top \mathbf{1}) = (1 + \tau)r$. Finally, $\tau(B) = \mathbf{1}^\top (c + \tau r) = \tau + \tau^2 = 0$ in $\mathbb{F}_2$. $\square$

2020 *Mathematics Subject Classification.* 37P25, 05C20, 15B33, 05A15.

Key words and phrases. finite dynamical system, directed graph, binary matrix, parity margin, transpose, fibre, functional graph.

Theorem 2.2 (complete temporal classification). *The image and complete recurrent set are both $\mathcal{H}_n$. Every state outside $\mathcal{H}_n$ has exact depth one. On $\mathcal{H}_n$,*

$$(3) \quad \Phi^2(A) = A + rr^\top + cc^\top, \quad \Phi^4(A) = A.$$

If $r \neq c$, then A has exact period four. If $r = c$, its period is one or two.

Proof. The last identity in (2) shows that $\text{im } \Phi \subseteq \mathcal{H}_n$. The reverse inclusion follows from the explicit preimage in theorem 3.1. On $\mathcal{H}_n$ the margins swap, so a second use of (1) yields the first identity in (3); this added matrix has zero row and column margins. Repeating the same two steps cancels it in characteristic two, proving the second identity. Thus every point of $\mathcal{H}_n$ is recurrent. Every odd-total point reaches it after one step and cannot already be recurrent, proving exact depth one.

If $r \neq c$, then $rr^\top \neq cc^\top$ because the diagonal of $vv^\top$ is v ; hence $\Phi^2(A) \neq A$, and the period is exactly four. If $r = c$, (3) gives $\Phi^2(A) = A$. $\square$

Here depth means the least entrance time into the recurrent set. On a finite map this equals the distance to the first periodic point. The unique matrix in $M_0(\mathbb{F}_2)$ may separately be declared fixed; all remaining formulas assume $n \geq 1$.

3. EVERY ONE-STEP FIBRE AND A SECOND ROUTE

Theorem 3.1 (codomain-wide fibre trichotomy). *For every $B \in M_n(\mathbb{F}_2)$,*

$$(4) \quad |\Phi^{-1}(B)| = \begin{cases} 0, & \tau(B) = 1, \\ 1, & \tau(B) = 0 \text{ and } c(B) \neq 0, \\ 2^{n-1} + 1, & \tau(B) = 0 \text{ and } c(B) = 0. \end{cases}$$

In particular, $|\text{im } \Phi| = 2^{n-1}$, and precisely $2^{n(n-1)}$ image targets have the large fibre.

Proof. If a preimage has row margin r , solving (1) for it gives $A = B^\top + rr^\top$. Requiring $A\mathbf{1} = r$ is equivalent to

$$(5) \quad r = c(B) + (\mathbf{1}^\top r)r.$$

For even-weight r , this forces $r = c(B)$ and exists exactly when $\tau(B) = 0$; it then supplies one preimage. For odd-weight r , (5) is equivalent to $c(B) = 0$. In that case all 2^{n-1} odd vectors work, in addition to the even solution $r = 0$. This proves (4) and also $\text{im } \Phi = \mathcal{H}_n$.

There are 2^{n-1} choices for each even-parity column, independently over n columns, so exactly $2^{n(n-1)}$ matrices have $c(B) = 0$. $\square$

The odd-total zero case in (4) is essential. For example, when $n = 1$, the target $[1]$ has nonzero column margin but no preimage.

Proposition 3.2 (factor-product route). *The update has the factorisation*

$$(6) \quad \Phi(A) = (I + r\mathbf{1}^\top)A^\top.$$

If $\tau = 0$, the left factor is the identity when $r = 0$ and an involutory transvection otherwise. If $\tau = 1$, it is an idempotent rank- $n - 1$ projection with kernel $\langle r \rangle$ and image $\mathbf{1}^\perp$. Moreover, for $\tau = 0$ the factor product independently gives

$$(7) \quad \Phi^2(A) = (I + c\mathbf{1}^\top)A(I + \mathbf{1}r^\top) = A + cc^\top + rr^\top,$$

and hence $\Phi^4(A) = A$.

Proof. Since $\mathbf{1}^\top A^\top = r^\top$, (6) is (1). Put $L_r = I + r\mathbf{1}^\top$. Direct multiplication gives $L_r^2 = I + \tau r\mathbf{1}^\top$, while

$$(8) \quad \mathbf{1}^\top L_r = (1 + \tau)\mathbf{1}^\top, \quad \det L_r = 1 + \tau.$$

Thus $L_r^2 = I$ when $\tau = 0$, with the stated identity/transvection distinction. When $\tau = 1$, $L_r^2 = L_r$ and

$$L_r x = 0 \iff x = r(\mathbf{1}^\top x).$$

Here $r \neq 0$, and $L_r r = 0$, so $\ker L_r = \langle r \rangle$. Rank-nullity gives rank $n - 1$; (8) places the image in the $(n - 1)$ -dimensional space $\mathbf{1}^\perp$, hence the two are equal.

It remains to obtain the temporal identity from the factors. Suppose $\tau = 0$ and put $B = L_r A^\top$. Direct multiplication of this product by $\mathbf{1}$ and of its transpose by $\mathbf{1}$ gives row and column margins c and r , respectively. Therefore

$$\Phi^2(A) = L_c B^\top = L_c A L_r^\top.$$

Expanding the two factors gives (7); its mixed term is $c(\mathbf{1}^\top A \mathbf{1})r^\top = 0$. Because r and c have even weight, the correction $cc^\top + rr^\top$ has zero row and column margins. The same factor-product calculation on $\Phi^2(A)$ adds that correction once more, proving $\Phi^4(A) = A$ in characteristic two. $\square$

Thus the factor route both identifies the singular odd-coset projection and recovers the even-coset temporal identities; the direct margin equation in [theorem 3.1](#) is still needed to distinguish individual target fibres.

4. COMPONENTS AND ZETA FUNCTION

Lemma 4.1 (prescribed margins). *For $u, v \in \mathbb{F}_2^n$, there is a matrix with row margin u and column margin v exactly when $\mathbf{1}^\top u = \mathbf{1}^\top v$. In that case there are $2^{(n-1)^2}$ such matrices.*

Proof. Choose the upper-left $(n-1) \times (n-1)$ entries freely. The prescribed row sums force the last entry of each of the first $n-1$ rows, and the prescribed column sums force the last entry of each of the first $n-1$ columns. The two requirements on the bottom-right corner agree precisely when the total parities agree. $\square$

Put

$$F_n = 2^{n(n-1)/2}, \quad C_{2,n} = \frac{2^{n(n-1)} - F_n}{2}, \quad C_{4,n} = \frac{2^{n^2-1} - 2^{n(n-1)}}{4}.$$

Theorem 4.2 (functional components). *The map has F_n fixed points, $C_{2,n}$ two-cycles, and $C_{4,n}$ four-cycles. It has 2^{n^2-1} depth-one states. With the usual finite-map convention [1],*

$$(9) \quad \zeta_{\Phi_n}(z) = (1-z)^{-F_n} (1-z^2)^{-C_{2,n}} (1-z^4)^{-C_{4,n}}.$$

Proof. A fixed point satisfies $A + A^\top = rr^\top$. Comparing diagonals forces $r = 0$; then A is symmetric with zero row sums. Its off-diagonal entries are arbitrary and its diagonal is forced by the row sums, giving dimension $n(n-1)/2$ and hence F_n fixed points.

By [lemma 4.1](#), for each of the 2^{n-1} even vectors r , there are $2^{(n-1)^2}$ recurrent matrices with margins (r, r) . Thus $2^{n(n-1)}$ recurrent points have period dividing two. Removing fixed points and dividing by two gives $C_{2,n}$. All remaining recurrent points have exact period four by [theorem 2.2](#), giving $C_{4,n}$. The odd coset has size 2^{n^2-1} and consists exactly of the depth-one points. Finally, the standard product of one factor $(1-z^k)^{-1}$ per k -cycle gives (9). $\square$

5. EXACT CONTROL AND LIMITATIONS

The accompanying deterministic verifier exhausts every one of the 2^{n^2} matrices for $1 \leq n \leq 4$. It checks both forms of the update, the quotient and fourth-iterate identities, exact depths and periods, all codomain fibres including zero fibres, the affine margin count, and every component formula. These bounded computations are counterexample searches, not proofs of the all- n statements above.

The result is specific to the binary field, square matrices, transpose, and the recomputed row-margin selector. Rectangular carriers, other fields, other bilinear forms, and static graph complementation require new proofs and owner gates. Literature screening was bounded and can establish a hit but not an absence. Accordingly we make no claim of novelty, priority, authorship, posting readiness, or external release.

REFERENCES

- [1] Michael Artin and Barry Mazur. On periodic points. *Annals of Mathematics*, 81(1):82–99, 1965. doi: 10.2307/1970384.
- [2] André Bouchet. Recognizing locally equivalent graphs. *Discrete Mathematics*, 114(1–3):75–86, 1993. doi: 10.1016/0012-365X(93)90357-Y.
- [3] Robert Brijder and Hendrik Jan Hoogeboom. The group structure of pivot and loop complementation on graphs and set systems. *European Journal of Combinatorics*, 32(8):1353–1367, 2011. doi: 10.1016/j.ejc.2011.03.002.
- [4] Alex Buchanan, Christopher Purcell, and Puck Rombach. Subgraph complementation and minimum rank. *Electronic Journal of Combinatorics*, 29(1):P1.38, 2022. doi: 10.37236/10325.
- [5] Yin Chen and Shan Ren. Modular matrix invariants under some transpose actions. *Finite Fields and Their Applications*, 113:102824, 2026. doi: 10.1016/j.ffa.2026.102824.
- [6] Ivo Koch, Nina Pardal, and Vinicius F. dos Santos. Exploring subgraph complementation to bounded degree graphs, 2025.
- [7] Lorenzo Traldi. On the linear algebra of local complementation. *Linear Algebra and its Applications*, 436(5):1072–1089, 2012. doi: 10.1016/j.laa.2011.06.048.